

Characterizing Vapor-Pressure Profiles for Ba-Mg Alloy

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ABSTRACT

This study investigates BaMg (20% Ba, 80% Mg) as a suitable alternative source of barium for ion trap quantum computing applications to address operation challenges associated with pure barium’s atmospheric reactivity. Under high vacuum conditions ($< 10^{-6}$ mbar), we characterized vapor pressure profiles for both BaMg alloy and pure barium using temperature-controlled heating and residual gas analysis. The onset of barium vapor was detected at temperatures of 530 – 600°C for both materials, with peak partial pressures of $2 - 2.5 \times 10^{-7}$ Torr. The BaMg alloy demonstrated successful barium vapor production without requiring inert atmosphere handling, eliminating the complexities of loading protocols such as dry-ice bath environments that would be required for pure barium. While the measured pressures fell below equilibrium predictions due to dynamic heating under continuous vacuum pumping, the results demonstrate that the BaMg alloy can provide the gaseous barium atoms needed for ion trap applications, all while offering significant operational advantages. These findings suggest that the BaMg alloy could serve as a suitable alternative source of barium in current ion trap quantum computing systems.

1. INTRODUCTION

Barium is commonly used in ion traps for quantum computing due to its simple electron structure, accessible transition wavelengths, and long-lived metastable states suitable for qubit operations. However, metallic barium’s high reactivity with atmospheric oxygen and water vapor leads to surface oxidation that reduces vapor production and creates inconsistent ion generation rates. Current protocols require handling pure barium in inert atmospheres through specialized procedures such as dry-ice baths that maintain CO₂ environments during loading. These stringent requirements increase system complexity and operational costs. To address these limitations, we investigate the potential of alternative barium sources such as a BaMg alloy. We hypothesize that the alloy matrix provides reduced oxidation while maintaining sufficient barium vapor pressure for ion trap applications. Thus, the implementation of alloys into current trapping architecture could eliminate the need for inert atmosphere handling, reducing system complexity, and provide more stable and reproducible ion loading rates compared to pure barium sources.

2. METHODS

2.1. *Materials*

The BaMg alloy used consists of a 20% barium and 80% magnesium (with fractional impurities of Fe, Al, Ni, Cu with 0.01-0.05%) [S. A. Materials \(2024\)](#). In addition, a reference control of pure barium was used as comparison, and was handled under inert atmosphere conditions to minimize surface oxidation.

2.2. *Experimental Apparatus*

The experiments were conducted using a four-way cross vacuum chamber. The chamber was pumped sequentially through stages: a mechanical pump reduced pressures to approximately 1 mbar, followed with a Pfeiffer turbo pump achieving 10^{-6} mbar, and finally a Varian MiniVac ion pump to reach pressures of 10^{-6} to 10^{-7} mbar. This sequence ensured that the ultra-high vacuum conditions were met to accurately profile the vapor pressure measurements of the barium.

To analyze the vapor composition within the chamber, a Stanford Research Systems RGA200 residual gas analyzer (RGA) was used. The RGA’s mass range of 0-200 amu enabled proper coverage for detecting barium at an atomic mass of 138. The RGA ionizer extended into the central cross-section of the vacuum chamber, allowing for the intercept of the vapor that emanated from the sample oven.

The sample oven consisted of a cylindrical chamber mounted on a conflat flange that contained integrated copper feedthroughs. The oven was connected across two opposing channels with an opening directed toward the RGA ionizer to ensure efficient vapor transfer. Heating of the oven was achieved through resistive heating using an external power supply that was connected to the same copper feedthroughs on the outside of the flange. The heating current would pass through the oven assembly, allowing controlled sample vaporization.

Temperature monitoring was conducted using a K-Type thermocouple (chromel-alumel) and was configured to minimize electrical interference from the heating current. The hot junction was formed by spot-welding the chromel and alumel wires together with one wire extended to make thermal contact with the backside of the oven. This differential measurement approach isolated the temperature signal from the heating current interference [T. G. Ballance et al. \(2018\)](#). The distance between the hot junction and the oven contact point was minimized to reduce thermal gradient errors. Both thermocouple wires were connected to other copper feedthroughs and brought to a cold junction at a thermocouple readout that converted the thermoelectric voltage to temperature. The introduction of a third dissimilar metal in the copper feedthroughs was negligible as the temperature of both feedthroughs were equivalent through atmospheric exposure [J. Wu \(2023\)](#).

2.3. Procedure

Once the sample was loaded into the system and the desired vacuum was achieved ($< 10^{-6}$), the power supply current was increased in 1A increments with time intervals determined by monitoring temperature stabilization. This timing protocol was established in early test runs, but was consistently applied across subsequent measurements to ensure thermal equilibrium was achieved at each current setting.

During each heating cycle, the thermocouple temperature, relative system pressure (measured as the turbo pump), and mass-resolved partial pressure measurements for signals at 24 amu and 138 amu were continuously monitored. The threshold for detections were defined as signals exceeding the noise floor of approximately 10^{-9} Torr.

3. RESULTS

The first set of results were recorded manually across all tests at each point amperage step, with an emphasis of comparing the vapor-pressure profiles to the temperature.

Figure 1 presents the temperature characterization for the BaMg alloy and pure barium samples and the pressure characterization for the vacuum system. Temperature measurements at each amperage step (Fig. 1A) show the thermal response of the thermocouple on multiple test runs. The pressure readings of the turbo pump (Fig. 1B) were monitored primarily for instrument protection, as both the RGA and the ion pump required pressures below 10^{-6} mbar for proper operation. These measurements also provided an indication of the general outgassing from the oven region. The barium vapor detection results (Fig. 1C) present the maximum partial pressure recorded at each amperage step.

The temperature profiles showed a systemic decrease in subsequent runs due to progressive clogging of the oven, reducing the material loading and the efficiency of the vapor production. This effect necessitated the replacement of the oven between the alloy and pure barium tests, including the thermocouple assembly. The irregular pure barium signal at approximately 320° C (Fig. 1C) was identified as an instrumental artifact, as the duration and magnitude were inconsistent with the known barium vapor pressure behavior. It likely resulted from the detection of interference species within the RGA's ± 0.5 amu resolution.

Figure 2 presents the time-resolved partial pressure measurements for magnesium (24 amu) and barium (138 amu) throughout complete heating cycles (300 K to 950K). The vertical dashed lines indicate 1A current increases. These continuous scans reveal the temperature thresholds for the onset of vapor production, demonstrating the correlation between heating current and mass-resolved vapor signals. Although the peaks of the pure barium and the BaMg alloy were displaced (Fig. 2B), the thermocouple temperature profile indicates that the emergence of peaks occurred at the same temperature threshold. The artifact previously observed in the pure barium scan (at 4A) is evident and was excluded from further analysis based on its inconsistency with other recorded measurements. Other artifacts observed through the scans, especially those of the magnesium, are believed to include impurities from organic residue and outgassing of oven materials, as the atomic mass of 24 ± 0.5 is common for compounds containing low-mass elements.

4. DISCUSSION

The comparable onset temperatures for barium vapor in both alloy and pure samples demonstrate that BaMg alloy can provide suitable vapor production while eliminating inert atmosphere handling requirements. The performance

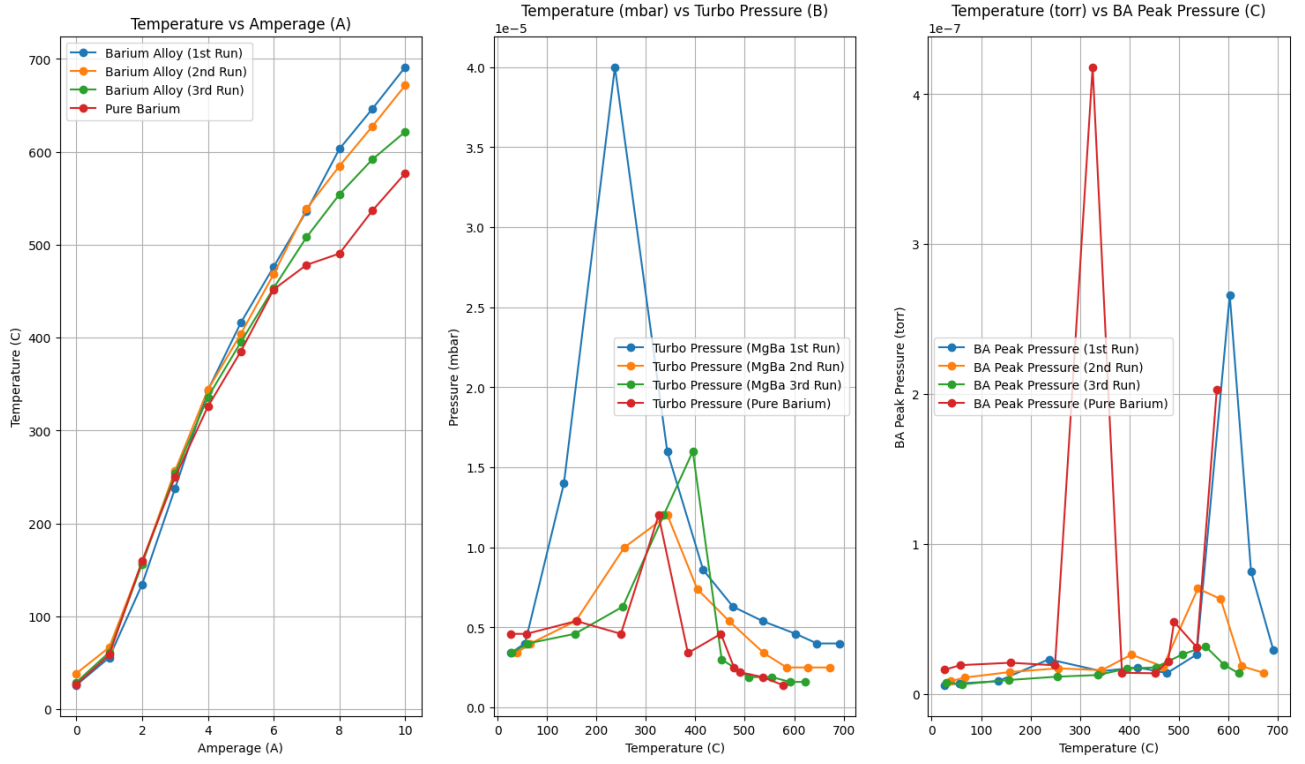


Figure 1. Fig. 1A compares the temperature stabilization at each amperage step across all test runs. Fig. 1B displays the maximum recorded turbo pressure reading for each amperage step (points). Fig. 1C highlights the maximum observed partial pressure measurement of the RGA vs the temperature.

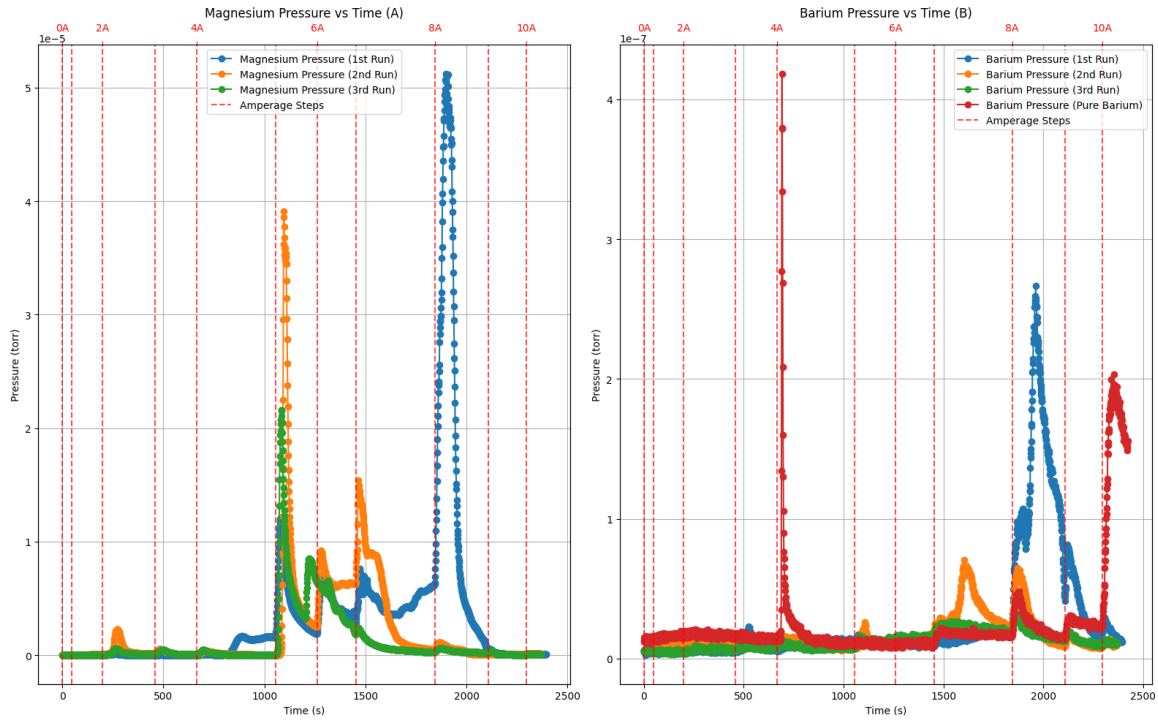


Figure 2. The two plots display the partial-pressure scans of magnesium (Fig. 2A) and barium (Fig. 2B) across the length of the scan. The vertical dashed lines indicate the time in which the amperage was increased by 1A.

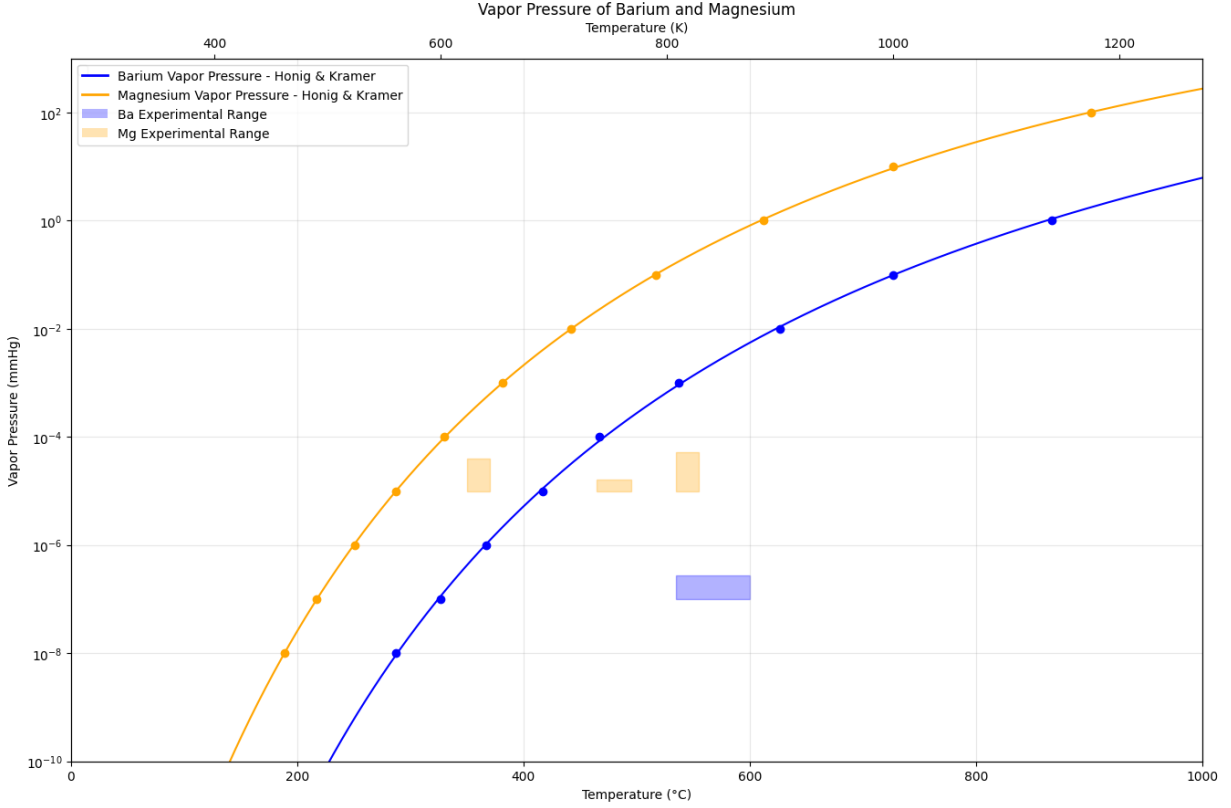


Figure 3. Using the vapor-pressure curves of both barium and magnesium, we highlight where recordings of $< 1e-5$ mmHg for magnesium and $< 1e-7$ mmHg for barium through vertical slices. The curves were derived by integrating the Gibbs-Helmholtz equation for phase equilibrium: $\log_{10} P = A/T + B \cdot \log_{10}(T) + C \cdot T + D$, where the constants (A, B, C, D) were fitted coefficients R. E. Honig (1962) & D. R. Stull & G. C. Sinke (1956).

trade-offs for alloy use regard the abundance of other materials (mainly magnesium) that fill the system. However, for trapping systems, in which only selective ions are confined to the desired cavity, the surplus material would not conflict with operations.

When compared to known literature (Fig. 3), our measured pressures fall significantly below theoretical predictions for the observed temperatures. The discrepancy lies in the system approach, where in the R. E. Honig (1962) system, the vapor accumulates until it reaches equilibrium pressure in a seal container, while we are constantly removing vapor through the vacuum pumps, thus never reaching an equilibrium pressure or steady-state partial pressures (step-function behavior). Our detected pressure ranges represent dynamic steady-state rather than true equilibrium conditions. This distinction doesn't signify a limitation but rather a realistic demonstration of vaporization under vacuum conditions similar to ion trap systems. As importantly, our results demonstrate that barium vapor is readily produced from the BaMg alloy at temperatures ranging $530 - 600^\circ \text{C}$, the same as that detected for pure barium. Thus validating the alloy for maintaining necessary thermal characteristics for vapor production while offering operational advantages in trapping architecture.

A major limitation of the results is the lack of repeated measurements; this is most apparent in testing of the alloy, where only one oven setup was repeated, with progressively lower partial pressures through ongoing tests. Further trials would have required new ovens for each test run. However, oven installation necessitated rebuilding the thermocouple circuit due to delicate wiring constraints. As seen between the temperature profiles of the BaMg alloy and the pure barium, replacing the circuit leads to different temperature profiles. This can be attributed to varying amounts of contact with the copper feedthroughs, changing the efficiency of heat transfer through the intermediate feedthroughs. Variations in hot junction positioning creates different thermal gradients. Uniform installation of the thermocouple circuit proved difficult with the precision required to spot-weld the thin wires at the two junctions.

Further testing of the pure barium sample would have significantly improved the comparison of barium signatures in creating a unified vapor-pressure profile and would have answered our belief that the artifact at 4A was due to error.

5. CONCLUSION

In conclusion, the observed vapor pressure peaks during heating represent successful material vaporization, precisely the behavior required for ion trap applications where gaseous barium atoms are needed for ionization and trapping. With peaks occurring at comparable temperatures for both materials, the BaMg alloy can serve as an effective barium source without compromising performance.

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